Assignment 02

Dataset Name: content_based_recommendation_dataset.csv

Submission Instructions:

- Complete this assignment using a Jupyter Notebook (.ipynb file).
- Write Python code for each question using Pandas and Matplotlib.
- Show the output of each question clearly.
- Upload your Jupyter Notebook file to GitHub.
- Submit your GitHub repository link on Google Classroom.

Q1. Load the dataset and display first 5 rows?

Q2. Display dataset shape and column names?

Q3. Check data types of all columns?

Q4. Identify missing values in each column?

Q5. Handle missing values?

Q6. Check and remove duplicate rows?

Q7. Generate summary statistics?

Q8. Calculate the average of 'Price of the product'?

Q9. Find max and min of 'Median purchasing price (in rupees)'?

Q10. Value counts of 'Brand of the product'?

Q11. Sort data by 'Price of the product'?

Q12. Filter data where rating > 4?

Q13. Create new column 'Price Difference'?

By using (Price of the product - Median Purchasing Price)

Q14. GroupBy average price by Gender?

Q15. Create histogram Price of the product?

Conclusion:

- Write 4 to 5 lines explaining:
- Which factor influences recommendations the most?
- What changes did you observe after cleaning the data?
- One interesting pattern you found.